



Figure 9: Option to enable 'CPU Idle Power Management' support

time on a device. You can also trigger the system to go into a *suspend* state by issuing the following command:

```
echo mem > /sys/power/state
```

The *suspend/resume* activity comes under the category of static power management. When in a suspended state, the system slows down or stops various clocks, removes power for devices or places them in a low-power state, thus lowering the core operating voltage and stopping the CPU. The power management subsystem in the 2.6 kernel implements a kernel API for *system suspend* and *resume* with limited support for choosing a specific *system suspend* state. The 2.6 *sysfs* filesystem exports interfaces that applications may use to suspend the system [3] and [12].

You can suspend to disk and resume (which is called 'hibernate') or you can suspend to RAM, as is done in many handheld devices. Figures 7 and 8 show how to enable these options during kernel configuration.

Device power management

It is a hardware or software mechanism that triggers a device into a low-power state after a period of inactivity. For example, a driver powers down the device because applications no longer hold an open reference to it. When the last open file descriptor for a device is closed, then the driver may suspend the device. Figure 9 shows the option to enable 'CPU Idle Power Management' support.

Dynamic power management

This refers to the dynamic control of clocks and power supplies, the adjustment of timer ticks, voltage and frequency scaling. This basically means power management when the device is active (i.e., when user applications are running). This can be done based on the processing load. Device drivers and operating systems can also contribute to the dynamic power management by reducing the speed of disks when the system is idling. Dynamic power management uses a combination of both user space and kernel space software.

In dynamic power management, you adjust power parameters on-the-fly when simultaneously making sure real-time requirements are satisfied. This uses recent improvements in hardware to scale clocking information and required voltages with reduced latency [13] and [14].

Dynamic power management focuses on quick application restart after standby, automating system parameters, where the power parameters are automatically selected based on CPU load, the number of high priority tasks existing, etc. Ultimately, more flexibility is available to the systems designer in constrained working environments such as the consumer electronics devices, in order to control the system parameters [3] and [15].



Note: We'll discuss real-time specifications and size reduction next month. **END**

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